

Network Guard: A Lightweight Intrusion and Data Leak Detection System

Project Synopsis Submitted

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MANIPAL ACADEMY OF HIGHER EDUCATION

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Of

Bachelor of Technology

in

Computer and Communication Engineering

by

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Objective:

The primary objective of this project is to design and develop "Network Guard," a lightweight, real-time monitoring tool that functions as both a mini Intrusion Detection System (IDS) and a Data Loss Prevention (DLP) system. The system aims to protect personal or small-scale networks by actively analyzing traffic to detect common threats and prevent the leakage of sensitive data.

Scope:

The scope of this project is to implement a system with the following core functionalities:

- **Intrusion Detection:**
 - **Port Scanning Detection:** The system will identify and alert when a single external IP address attempts to connect to multiple ports in rapid succession.
 - **ARP Spoofing Detection:** It will monitor Address Resolution Protocol (ARP) packets to detect potential man-in-the-middle attacks on the local network.
 - **Traffic Anomaly Detection:** The tool will flag unusual and sudden spikes in network traffic volume, which could signify a denial-of-service attack or malware activity.
- **Data Leak Detection:**
 - The system will scan the payload of outgoing network packets to identify and flag sensitive information using regular expressions.
 - Patterns to be detected include:
 - Email Addresses
 - Credit Card Numbers
 - Phone Numbers
 - Location Patterns (e.g., GPS coordinates)
- **Alerting and Logging:**
 - Upon detecting a threat, the system will generate an immediate alert on the console.
 - All detected events will be logged with a timestamp into a CSV file for subsequent analysis.

Need for the Application:

In today's interconnected digital world, network security is critically important. Personal and small-scale networks often lack the sophisticated and expensive security infrastructure of large enterprises, making them vulnerable to common cyber threats. Furthermore, the risk of accidentally leaking sensitive personal or financial information is a significant concern for every user. This project addresses the need for a simple, lightweight, and accessible tool that provides an essential layer of security. It empowers users to monitor their own networks for malicious reconnaissance, active attacks, and data exfiltration without requiring significant computational resources or specialized knowledge.

Project Description:

"Network Guard" is a real-time monitoring tool designed to secure a network by combining the functionalities of an Intrusion Detection System (IDS) and a Data Loss Prevention (DLP) tool. The system actively listens to network traffic, analyzing it on the fly to provide immediate protection.

The **problem statement** is the dual risk of unauthorized network intrusions and inadvertent data leakage from a personal or small-scale network.

The implemented functionalities will address this problem as follows:

1. **Intrusion Detection:** To counter external and internal threats, the system will monitor for reconnaissance activities like **port scanning**, detect man-in-the-middle attacks by identifying **ARP spoofing**, and recognize potential malware or denial-of-service attacks by flagging **traffic anomalies**.
2. **Data Leak Detection:** To prevent sensitive information from leaving the network, the application will inspect the content of outgoing data packets. Using regular expression matching, it will identify and flag patterns corresponding to **email addresses, credit card numbers, phone numbers, and location data**.
3. **Alerting and Logging:** When any of the above threats are identified, the system will instantly print a descriptive alert to the console and record the event, along with a timestamp, in a CSV log file for auditing and review.

Hardware Requirements:

Laptop, Network Interface Card (NIC)

Software Requirements:

Python, Scapy, Flask, HTML, Tailwind

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